

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Österreich (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

Mühlbauer GmbH & Co. KG

Frau Tugce Zabun

Josef-Mühlbauer-Platz 1

93426 Roding

Deutschland

Subject: Application for Application Engineer Image Processing (m/f/d)

Dear Ms. Zabun,

As a physics engineer specialized in high-precision metrology and industrial machine vision, I am applying for the Application Engineer position at your Roding site. Your focus on hardware and optical development over pure software programming perfectly aligns with my passion for building and optimizing physical optical systems. I have closely followed Mühlbauer's innovative contributions to automation and smart factory solutions, and I am eager to bring my hands-on expertise in advanced optics to your dynamic engineering team.

In my previous industrial role as a Machine Vision System Engineer at INTECOL S.A.S., I successfully designed and deployed end-to-end vision pipelines for harsh manufacturing environments. Using HALCON and smart cameras, I implemented high-speed beverage bottle inspection systems, carefully developing custom illumination strategies—including specifically tailored dark-field and coaxial lighting—to reliably detect liquid levels and microscopic printing defects. Furthermore, I commissioned vision systems for auto-part recognition at a Renault subdivision, integrating these sensors directly with robotic arms for flame-surface treatment processes. My rigorous approach to Quality Assurance, including systematic Gage R&R assessments and feasibility studies, ensured the 24/7 reliability of these crucial vision components on the active factory floor.

During my PhD at the Medical University of Innsbruck, I have deepened my technical expertise by building automated, high-resolution optical setups from scratch. I led the end-to-end assembly of a custom multiphoton microscope, which required the precise opto-mechanical adjustment of galvo scanners, high-power laser sources, and photomultiplier tubes to achieve sub-micron imaging resolution. Beyond the hardware setup, I programmed a robust Python and C++ control stack to seamlessly synchronize high-speed scientific cameras with active wavefront correctors for real-time sample characterization. This role also required me to actively manage technical specification sheets and rigorously qualify precision hardware from leading vendors such as Thorlabs and Edmund Optics.

This combination of hands-on experience in precise system alignment, hardware validation, and industrial automation enables me to bridge the gap between complex optical research and reliable industrial deployment. With my strong background in QA processes and an analytical mindset geared towards complex problem-solving, I am confident I can help optimize Mühlbauer's vision systems for your clients.

I am fully integrated into the DACH work culture, hold a B2 proficiency in German, and, as requested by the position, am highly willing to support international service assignments to assist your global operations. I look forward to the opportunity to discuss how my technical expertise, cross-functional communication skills, and adaptable nature can be a valuable asset to your team.

Sincerely,

Juan David Muñoz-Bolaños

Application Engineer Image Processing

Medizinische Universität Innsbruck

Juan David Muñoz-Bolaños

Application Engineer Image Processing

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+43 676 9115145

juan.munoz@i-med.ac.at | [LinkedIn Profile](#)



Summary

Dedicated R&D Optical Engineer with a robust background in combining **High-Precision Optical Metrology** with **Industrial Machine Vision** systems. Expert in developing advanced illumination concepts, real-time image processing algorithms (HALCON/Python), and hardware-software integration for automated inspection units. Proven problem-solving skills with a track record of bridging the gap between complex optical research and reliable industrial deployment.

Professional Experience

INTECOL S.A.S.

Junior Machine Vision System Engineer

Medellín, Colombia

Mar 2018 – Mar 2019

Vision Pipeline Engineering: Designed and implemented Halcon-based beverage bottle inspection systems; developed custom illumination strategies to reliably detect liquid levels and printing defects.

Industrial Integration: Commissioned vision systems using smart cameras and custom illumination for auto-part recognition at a car subdivision of Renault; integrated vision sensors with robotic arms for flame-surface treatment processes.

QA & Feasibility: Performed systematic Gage R&R assessments and feasibility studies for new industrial clients, ensuring 24/7 reliability of vision components on the floor.

Medical University of Innsbruck

PhD Researcher – Optical Metrology & Automation

Innsbruck, Austria

Jan 2022 – Present

System Build & Justage: Led the end-to-end assembly of a custom multiphoton microscope; adjusted galvo scanners, laser sources, and PMTs to achieved sub-micron resolution.

Adaptive Optics Control: Programmed a Python/C++ control stack to synchronize scientific cameras with active wavefront correctors for real-time sample characterization.

Vendor & Component QM: Managed technical specification sheets and qualified precision hardware from Thorlabs and Edmund Optics as part of system procurement.

Leibniz Institute of Photonic Technology (IPHT)

R&D Researcher (Photonics & Data)

Jena, Germany

Jun 2020 – Dec 2021

Instrumentation Design: Prototyping of a dispersed fiber-probe Raman spectrometer; handled optical layout justage and hardware-software calibration.

Data Evaluation Hooks: Integrated computational data analysis (Python ML) with hardware sensors to optimize measurement throughput and validation.

Technical Competencies

Machine Vision Hardware: Mastering of CMOS/CCD camera integration, lens selection (telecentric, macro), and custom illumination design (Dark-field, Bright-field, coaxial).

Image Processing Libraries: Advanced proficiency in Halcon and Cognex VisionPro for real-time inspection, part recognition, and measurement. Familiar with OpenCV and Python-based vision pipelines.

System Alignment & Justage: Hands-on experience in the precise alignment of complex opto-mechanical setups (Laser sources, scanners, detectors). Expert in qualifying components from market-leading vendors.

Automation & Integration: Skilled in automating optical systems with Python, C++, and LabVIEW.

Experienced in ensuring Gage R&R and measurement reliability in harsh industrial environments.

R&D Methodology: Experience bringing systems from concept to series prototypes, including ZEMAX modeling, procurement management, and technical documentation.

Education

Medical University of Innsbruck

Ph.D. in Adaptive Optics & Microscopy

Austria

Jan 2022 – Present

Friedrich Schiller University Jena

M.Sc. in Photonics

Germany

2019 – 2021

University of Cauca

B.Sc. in Physics Engineering

Colombia

2012 – 2018

Publications & awards

Awards: ASML & ZEISS Summer Schools participant; COLFUTURO Grant 2026 awardee.

Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).

Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).

Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

Languages

English (Professional),

German (B2 - Working Proficiency),

Spanish (Native)

Soft skills

Collaboration & Communication: Proven ability to work in cross-functional R&D teams and communicate complex technical concepts effectively.

Problem Solving: Strong analytical mindset with a track record of troubleshooting and resolving hardware-software integration challenges.

Adaptability: Quick learner, highly adaptable to new technologies, and experienced in international industry environments.

Hobbies

Mountain Biking, Dancing Bachata, Learning Harmonica.

References

Prof. Monika Ritsch-Marte

Head of Institute of Biomedical Physics

monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Kraft

Spectroscopy Group Leader

christoph.kraft@leibniz-ipht.de

Leibniz Institute of Photonic Technology

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

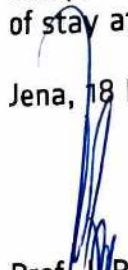
Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garces
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

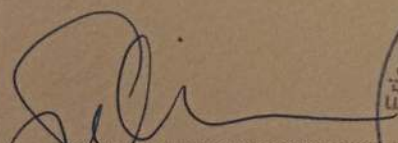
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy



INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532

INTECOL

DEN RHYTHMUS DER WELT OPTIMIEREN

Medellín, den 4. Februar 2019

AN ALLE, DIE ES BETRIFFT

Integración Tecnológica Industrial Intecol S.A.S. mit der Steuernummer (NIT) 811.041.410-4 bescheinigt, dass Herr JUAN DAVID MUÑOZ BOLAÑOS, identifiziert mit dem Personalausweis Nr. 1.061.772.278, seit dem 3. Juli 2018 in diesem Unternehmen beschäftigt ist. Er hat einen unbefristeten Arbeitsvertrag und ist in der Position als Junior-Projektingenieur tätig;

er bezieht ein Gehalt von einer Million fünfhunderttausend Pesos (\$1.500.000) zuzüglich dreihunderttausend Pesos (\$300.000) Mobilitätszulage.

Für weitere Informationen stehe ich Ihnen gerne unter der Telefonnummer 316 3322 zur Verfügung.

Wir danken für Ihre Aufmerksamkeit,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Verwaltungs- und Finanzdirektorin

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellín Calle 42 #63c-82
PBX. (574) 316 3322 Handy: (57) 301 430 2532

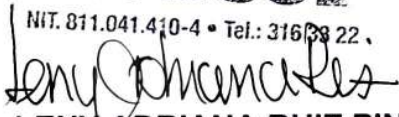
Medellín, 4 de Febrero de 2019

A QUIEN PUEDA INTERESAR

Integración Tecnológica Industrial Intecol S.A.S. con NIT.811.041.410-4 certifica que el señor **JUAN DAVID MUÑOZ BOLAÑOS**, identificado con cédula de ciudadanía N°1.061.772.278, labora en esta compañía desde el 3 de Julio de 2018, con un contrato a término indefinido, desempeñando el cargo de Ingeniero de proyectos Junior; devengando un salario Un millón quinientos mil pesos ml (\$1.500.000) más trescientos mil pesos ml (\$300.000) de movilidad.

Cualquier información adicional con gusto la atenderé en el teléfono 316 3322.

Agradecemos su atención,

INTECOL
NIT. 811.041.410-4 • Tel.: 316 3322,

LENY ADRIANA RUIZ PINO
Directora Administrativa y Financiera
Integración Tecnológica Industrial Intecol S.A.S.

Passaporto № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

10 DIC/DEC 2028

G. ANTIOQUIA

Firma del titular / Titular y representante

Quar Family

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